From the Wall-Tap

A Standalone Derivation of Resolution Geometry: Records, Descent, Order, Topology, Gluing, Fisher Geometry, Development and the Sign, with Every Assumption Exposed and Established Names Given Only Afterwards

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Abstract

Earlier RG papers imported many established theorems. That made each step safe, but it hid the unity of the structure and its exact gaps. Here the order is reversed. We start from one experiment: tap a wall, wait, record. Everything is derived with complete proofs from that experiment and from assumptions that are stated as they enter. No named characterization theorem is used as a premise. The mathematics admitted throughout is finite sums, real numbers and limits, and finite-dimensional linear algebra. Only after each derivation do we identify the result with its established name (Čencov, Blackwell, Čech, Cramér–Rao and others).

The chain.

- Records give stochastic processing, with composition as matrix multiplication.
- Exact laws give the observational quotient and descent, and the predictive quotient, the coarsest exact state.
- Simulation gives monotone decision risk. Universally reversible refinements are exactly disjoint splittings.
- Finite records give a topology whose open sets are the confirmable propositions. It is Hausdorff on the quotient.
- Overlapping observers give gluing: a global description exists iff every cycle's holonomy is trivial.
- Under declared regularity, the unique splitting-invariant positive quadratic geometry is c·Fisher. Approximate invariance gives it within $e^{\pm 2\varepsilon}$.
- Contraction comes with an exact sum-of-squares deficit, and from it the Hessian selection, the regular error bound, monotone development and the confirmation identity.
- A Lorentz sign appears only by an added radial declaration.

Obstructions. Five obstructions show which further claims are *not* forced by tapping: unperformed protocols, finite records, mediators, smoothness and signature. Eleven exact witness groups check the identities in rational arithmetic.

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1 The starting object and the rules of this paper

The experiment. A history is a finite string $(a_1, x_1, \ldots, a_n, x_n)$ of interventions a_k (tap, wait, move a detector) and recorded outcomes x_k from finite alphabets. A *protocol* chooses the next intervention, possibly depending on earlier outcomes.

Rules.

(R1) Every assumption is stated where it enters, and is listed in the ledger (Section 12).

(R2) The mathematics admitted throughout consists of finite sums, the real numbers (density of rationals, limits, continuity), and finite-dimensional linear algebra: polarization, Sylvester's law of inertia, and, in one marked place, the finite separation lemma.

(R3) Established names appear only in Section 11, after the proofs.

2 Records and processing

Assumption A (Stable statistics)**.** For a fixed preparation θ and protocol p, the relative frequencies N_x/N of each outcome converge as the number of repetitions $N \to \infty$. Call the limit $P_\theta^{(p)}(x)$.

Proposition 2.1 (Normalization and additivity)**.** $P_\theta^{(p)}(x) \geq 0$, $\sum_x P_\theta^{(p)}(x) = 1$, *and for disjoint groups of outcomes the limit of the group's frequency is the sum of the limits.*

Proof. Frequencies are nonnegative and sum to one at each N, and finite sums commute with limits. □

The *existence* of the limits is assumed, not derived. This is the first place where "forced" must read "assumed".

Assumption B (Record-only processing)**.** A processing step converts a stored outcome x into a reported outcome y with probabilities $K(y \mid x)$ that do not depend on θ.

Lemma 2.2 (Processing is a stochastic matrix; composition is multiplication)**.** *The reported law is* $(KP_\theta)(y) = \sum_x K(y \mid x)P_\theta(x)$*. Two successive steps give the matrix product* LK*, and composition is associative.*

Proof. Partition trials by their stored outcome, and use additivity: $(LK)(z \mid x) = \sum_y L(z \mid y)K(y \mid x)$. Associativity holds because finite sums can be rearranged. □

A second physical tap may change the room. It is *not* automatically a processing of the first tap's record.

3 Descent, refinement and the predictive quotient

Definition 3.1 (Profile, observational equivalence)**.** For a set Θ of candidate descriptions and admitted protocols $\mathcal{P}$, the profile is $\Phi(\theta) = (P_\theta^{(p)})_{p\in\mathcal{P}}$. Write $\theta \sim \theta'$ iff $\Phi(\theta) = \Phi(\theta')$. A quantity Q is *determined* if it is a function of the profile.

Theorem 3.2 (Descent)**.** *Q is determined iff Q is constant on every class of $\sim$.*

Proof. If $Q = \bar{Q} \circ \Phi$, equal profiles give equal values. Conversely, if Q is constant on classes, define $\bar{Q}(\Phi(\theta)) = Q(\theta)$. This is well defined because constancy on classes makes the value independent of the representative. □

Proposition 3.3 (Refinement)**.** *If $\mathcal{P} \subseteq \mathcal{P}'$, then $\theta \sim_{\mathcal{P}'} \theta'$ implies $\theta \sim_{\mathcal{P}} \theta'$. More protocols can split classes and never merge them.*

Assumption C (Causal consistency)**.** The probability of an already recorded string does not depend on interventions chosen later.

Definition 3.4 (Predictive equivalence)**.** Histories h, h' are predictively equivalent, $h \approx h'$, if every admitted continuation protocol has the same law of future records after h and after h'. The class $[h]$ is the *predictive state*.

Theorem 3.5 (Predictive update)**.** *If $h \approx h'$ and (a, x) has positive probability after h, then $hax \approx h'ax$. Hence the update $[h] \mapsto [hax]$ is well defined.*

Proof. Equivalence gives equal probabilities of x under a after h and after h', and equal joint probabilities of x followed by any future record string. Assumption C makes these joint probabilities meaningful. Divide by the common positive probability of x. □

Proposition 3.6 (Minimality)**.** *Any state description from which all continuation laws can be computed refines $\approx$. So $[h]$ is a function of it, and the predictive quotient is the coarsest exact state.*

Proof. Equal descriptions give equal continuation laws, hence equivalent histories. □

Nothing here guarantees that the predictive state is finite-dimensional, smooth, spatial, or governed by a differential equation.

4 Order: simulation and reversible refinement

Theorem 4.1 (Monotone decision risk)**.** *For a prior π on finite Θ, actions $\mathcal{A}$ and losses $L(\theta, a)$, let $r(E) = \min_\delta \sum_\theta \pi_\theta \sum_x P_\theta(x) \sum_a \delta(a \mid x) L(\theta, a)$. Then for every processing K,*

$$r(KE) \geq r(E).$$

Proof. A rule δ on processed records defines the rule $\delta'(a \mid x) = \sum_y \delta(a \mid y) K(y \mid x)$ on original records. Substitution shows that the two have equal expected loss. The minimum over the larger set of rules cannot be larger. □

Corollary 4.2 (Equality under recovery)**.** *If a processing R (independent of θ) satisfies $RKP_\theta = P_\theta$ for all θ, then $r(KE) = r(E)$ for every decision problem.*

Theorem 4.3 (Universally reversible refinements)**.** *A stochastic K has a stochastic left inverse L ($LK = I$ on all input laws) iff, after deleting unused outputs, its columns have pairwise disjoint supports. That is, it splits each input symbol into its own group of outputs.*

Proof. Only if. $1 = (LK)_{xx} = \sum_y L(x \mid y) K(y \mid x)$ is an average of numbers $L(x \mid y) \leq 1$ with weights $K(y \mid x)$. So $L(x \mid y) = 1$ wherever $K(y \mid x) > 0$. If some y had $K(y \mid x) > 0$ and $K(y \mid x') > 0$ with $x \neq x'$, the column $L(\cdot \mid y)$ would contain two ones.

If. Group each output back to its unique source. Witness (E2): over the 225 quarter-grid maps from 2 inputs to 3 outputs, a left inverse exists exactly for the 24 maps with disjoint supports. □

Theorem 4.4 (Converse comparison)**.** *Admit the finite separation lemma (a closed convex cone and a point outside it can be separated by a hyperplane). If $r(F) \geq r(E)$ for every finite decision problem, then $F = KE$ for some processing K.*

Proof. Represent E by its posterior law $\mu_E = \sum_x m_x \delta_{b_x}$, where $m_x = \sum_\theta \pi_\theta P_\theta(x)$ and $b_x(\theta) = \pi_\theta P_\theta(x)/m_x$. Then:

1. Processing averages posteriors: $b_Y = \mathbb{E}[b_X \mid Y]$, by Bayes' rule and Lemma 2.2.
2. Decision values are integrals of convex piecewise-linear functions of the posterior, and every such function arises from some decision problem.
3. If the transport system $\sum_j \gamma_{ij} = m_i$, $\sum_i \gamma_{ij} b_i = n_j c_j$, $\gamma \geq 0$ were infeasible, the separation lemma would give a convex piecewise-linear Φ with $\int \Phi \, d\mu_E < \int \Phi \, d\mu_F$, contradicting the hypothesis.
4. Set $K(j \mid i) = \gamma_{ij}/m_i$.

□

5 Topology from finite records

Assumption D (Independent repetition)**.** Runs of a protocol on a fixed preparation are independent and identically distributed.

Lemma 5.1 (Hybrid bound)**.** *For laws P, Q on a finite alphabet and every event A of n records, $|P^{\otimes n}(A) - Q^{\otimes n}(A)| \leq n \ \mathrm{TV}(P, Q)$.*

Proof. Telescope: $P^{\otimes n} - Q^{\otimes n} = \sum_{k=1}^{n} P^{\otimes(k-1)} \otimes (P - Q) \otimes Q^{\otimes(n-k)}$. In the k-th term, fix the other coordinates. The section of A is an event of one record, and $|(P - Q)(\cdot)| \leq \mathrm{TV}$ on it. Integrating over the other coordinates keeps the bound. Witness: E6, all events for $n \leq 3$. □

Lemma 5.2 (An elementary verifier)**.** *For one binary coordinate with frequency f_n after n runs, $P(|f_n - p| \geq r) \leq 1/(4nr^2)$. With $r_n = (4n\alpha)^{-1/2}$, the rule "accept U iff $[f_n - r_n, f_n + r_n] \subseteq U$" has false acceptance at most α outside U, and acceptance tending to 1 inside an open U. For m outcomes, use level α/m per coordinate.*

Proof. $\operatorname{Var} f_n = p(1-p)/n \leq 1/(4n)$, and Chebyshev's inequality follows from $\mathbb{E}(f_n - p)^2 \geq r^2 P(|f_n - p| \geq r)$. Witness: E7, with $n = 400$ and $U = (0.3, 0.7)$; the worst false acceptance is 1.2×10^{-6}, and acceptance at 0.5 is 0.9996. □

Definition 5.3 (Confirmable proposition)**.** A set $A \subseteq \Theta$ is *confirmable* if for every $\alpha > 0$ there are finite-record rules with false acceptance at most α outside A and acceptance tending to 1 at every point of A.

Theorem 5.4 (Topology of confirmation)**.** *Let countably many protocols with finite alphabets be admitted, under Assumptions A and D.*

(a) The confirmable sets are exactly the preimages under Φ of open sets of the product of simplices. They form a topology.

(b) That topology is Hausdorff on the quotient $\Theta/\!\sim$.

(c) Two descriptions are topologically indistinguishable iff they are observationally equivalent.

Proof. (a) *Necessity.* Suppose A is confirmable at θ, with acceptance above $\alpha + \delta$ at some n. By Lemma 5.1, every θ' whose laws on the finitely many protocols used are within δ/n also has acceptance above α, so $\theta' \in A$. Thus A contains a basic neighbourhood.

Sufficiency. Basic open sets are confirmed by Lemma 5.2. Finite intersections are confirmed by splitting records and requiring both acceptances. Countable unions are confirmed by running the i-th verifier at level $\alpha 2^{-i}$ on its own block, with the number of blocks growing. Every open set of a countable product of simplices is a countable union of basic ones.

(b) Distinct profiles differ in some coordinate. Choose disjoint intervals around the two values.

(c) Preimage topologies identify exactly the points with equal images. □

6 Gluing observers

Definition 6.1 (Overlap system)**.** Observers $i \in V$ each hold a local description s_i on a region. An overlap graph has an edge ij when the regions overlap. Conversions $g_{ij} \in G$, for a group G acting on descriptions, satisfy $s_j = g_{ij} \cdot s_i$ on the overlap, with $g_{ji} = g_{ij}^{-1}$.

Lemma 6.2 (Cocycle on triple overlaps)**.** *If s_i, s_j, s_k exist on a common triple overlap and G acts freely, then $g_{jk} g_{ij} = g_{ik}$ there.*

Proof. $s_k = g_{jk} g_{ij} s_i = g_{ik} s_i$, and freeness cancels s_i. □

Theorem 6.3 (Gluing)**.** *On a connected overlap graph, the following are equivalent:*

(i) there are $h_i \in G$ with $g_{ij} = h_j h_i^{-1}$ for every edge (a global description);

(ii) the holonomy $g_{i_k i_{k-1}} \cdots g_{i_2 i_1} g_{i_1 i_k}$ is the identity around every cycle.

The global description is then unique up to one global element of G: a null.

Proof. (i)⇒(ii): the product telescopes to the identity.

(ii)⇒(i): choose a spanning tree and a root, and set $h_{\text{root}} = e$. Define $h_j = g_{ij} h_i$ along tree edges. A non-tree edge closes a cycle with tree edges, so its conversion agrees with the tree product exactly when that cycle's holonomy is trivial.

Uniqueness: two potentials differ by $h'_i h_i^{-1}$, which is constant along edges and hence constant on the connected graph.

Witness: E8. Over $\mathbb{Z}$, a triangle with holonomy 0 glues and one with holonomy 1 does not. Over $\mathbb{Z}_2$, the Möbius triangle with holonomy -1 does not glue. □

Corollary 6.4 (Pairwise agreement is not global closure)**.** *Every edge may be a perfect pairwise agreement while some cycle has nontrivial holonomy. Then no global description exists, although every local one is valid.*

7 Geometry: the forced quadratic form

Assumption E (Regular universal geometry)**.** For every $n \geq 2$ and every positive law p on n symbols, there is a positive definite bilinear form g_p on $T_n = \{v \in \mathbb{R}^n : \sum v_i = 0\}$, with these properties:

(G1) it is continuous in p;

(G2) it is invariant under relabeling of symbols;

(G3) it is invariant under every disjoint splitting (Theorem 4.3): $g_{Kp}(Kv, Kv) = g_p(v, v)$;

(G4) it forms one family across all n.

These are representation and regularity requirements. They are not consequences of one tap, and an apparatus need not realize the whole simplex. Requiring the geometry on all finite laws is an extension beyond the accessible family.

Theorem 7.1 (Forced geometry)**.** *Under Assumption E, there is a unique constant $c > 0$ such that*

$$g_p(u, v) = c \sum_i \frac{u_i v_i}{p_i}$$

for every n and every positive p.

Proof. (A) Uniform laws. Extend g at $u_n = (1/n, \ldots)$ to $\mathbb{R}^n$ by projecting onto T_n. The extension is permutation-invariant, so its matrix has one diagonal value a and one off-diagonal value b. On T_n, $v^\top G w = (a-b)\langle v, w\rangle + b(\sum v)(\sum w) = (a-b)\langle v, w\rangle$. So $g_{u_n} = c_n \langle \cdot, \cdot \rangle$ with $c_n > 0$.

(B) Sizes. Split each symbol into k equal labels. The uniform law goes to the uniform law, and v goes to v_i/k repeated k times, so $|Sv|^2 = |v|^2/k$ (E3). Invariance gives $c_{nk}/k = c_n$. Put $a_n = c_n/n$; then $a_{nk} = a_n$, so $a_n = a_{nm} = a_m$ for all n, m. Hence $c_n = nc$.

(C) Rational laws. For $p_i = k_i/N$, split symbol i into k_i equal labels. This gives the uniform law on N symbols, and u goes to u_i/k_i repeated k_i times. So $g_p(u, u) = Nc \sum_i k_i (u_i/k_i)^2 = c \sum_i u_i^2/p_i$ (E4).

(D) All positive laws. Rational positive laws are dense, so (G1) extends the formula. Polarization gives the bilinear form. □

The constant c is a unit of information. The tap does not fix it.

Theorem 7.2 (Non-ideal version)**.** *Replace (G3) by approximate invariance,* $g_{Kp}(Kv, Kv) \in [e^{-\varepsilon}, e^{\varepsilon}]\, g_p(v, v)$*, keeping exact relabeling invariance. Then there is* $c > 0$ *with*

$$g_p(u, u) \in [e^{-2\varepsilon}, e^{2\varepsilon}]\; c \sum_i u_i^2/p_i$$

for all p*.*

Proof. Step (A) is unchanged. Step (B) gives $a_{nk} \in e^{\pm\varepsilon} a_n$, hence $a_n/a_m \in e^{\pm 2\varepsilon}$ through a_{nm}. So all a_n lie in an interval whose endpoints have ratio at most $e^{2\varepsilon}$. Take c to be its geometric midpoint, so that $a_N \in e^{\pm\varepsilon} c$. Step (C) contributes one more factor $e^{\pm\varepsilon}$, and continuity extends the bound. □

Theorem 7.3 (Contraction with an exact deficit)**.** *Write* $F_p(u) = \sum_x u_x^2/p_x$*. For every processing* K*, summing over outputs with* $(Kp)_y > 0$*,*

$$F_p(u) - F_{Kp}(Ku) = \sum_{x,y} p_x K(y \mid x) \Big(\frac{u_x}{p_x} - \frac{(Ku)_y}{(Kp)_y} \Big)^2 \;\geq 0.$$

Equality holds iff u_x/p_x *is constant on the inputs feeding each output.*

Proof. Expand the square and use $\sum_y K(y \mid x) = 1$, $\sum_x K(y \mid x) u_x = (Ku)_y$ and $\sum_x p_x K(y \mid x) = (Kp)_y$. The three terms are $F_p(u)$, $-2F_{Kp}(Ku)$ and $+F_{Kp}(Ku)$. Witness: E1, 81 exact rational cases. □

Obstruction 7.4 (Local is not global). $p_\theta = (\frac{1}{2} + \theta^2, \frac{1}{2} - \theta^2)$ has $F = 0$ at $\theta = 0$, yet different laws for every $\theta \neq 0$. A constant record also has $F = 0$ there, but cannot recover the family. A zero local deficit is not global recoverability.

8 Downstream consequences

Theorem 8.1 (Hessian selection)**.** *Let* $D(p, q) \geq 0$ *satisfy* $D(p, p) = 0$*, be* C^2 *near the diagonal, be invariant under disjoint splittings, and have a positive definite, continuous diagonal Hessian* $h_p(u) = \partial_t^2 D(p + tu, p)|_0$*. Then* $h_p = cF_p$*. The finite values of* D *are* not *selected.*

Proof. Splitting invariance of D passes to h. Apply Theorem 7.1.

Non-uniqueness. $\sum p \log(p/q)$ and $\frac{1}{2}\sum (p-q)^2/q$ have the same Hessian and different finite values. A positive D may also start at fourth order, so nondegeneracy is an extra condition. □

Theorem 8.2 (Regular error bound)**.** *Assume a smooth positive model* p_θ*, the score* $s = \partial_\theta \log p_\theta$ *and an unbiased estimator* T*. Then* $\operatorname{Var} T \geq 1/F(\theta)$*, where* $F = \mathbb{E}s^2$*.*

Proof. $\mathbb{E}s = 0$ and $\mathbb{E}[Ts] = 1$, by differentiating $\mathbb{E}T = \theta$. Then $0 \leq \mathbb{E}(T - \theta - \lambda s)^2$ for all λ, and the minimum at $\lambda = 1/F$ gives the bound. Witness: E9, where the Bernoulli record attains it. □

Theorem 8.3 (Monotone development)**.** *For accumulated records* $X_{1:t}$*, deleting the later part is a processing. Therefore optimal decision risk is nonincreasing in* t *(Theorem 4.1). For regular models, Fisher information is nondecreasing in* t*, with increments equal to the deficit of Theorem 7.3. No independence of increments is needed.*

Theorem 8.4 (Confirmation)**.** *For two record laws* P_0, P_1 *with equal priors, the optimal error is* $\frac{1}{2}(1 - \mathrm{TV}(P_0, P_1))$*. If the laws coincide up to time* τ*, then no rule based on records up to* τ *does better than* $\frac{1}{2}$*, and the confirmation time* $t_\varepsilon = \inf\{t : \frac{1}{2}(1 - \mathrm{TV}_t) \leq \varepsilon\}$ *satisfies* $t_\varepsilon \geq \tau$*.*

Proof. The error of a test φ is $\frac{1}{2}[\sum P_0\varphi + \sum P_1(1-\varphi)]$, which is minimized by $\varphi = \mathbf{1}[P_1 > P_0]$, giving $\frac{1}{2}(1 - \sum(P_1 - P_0)_+)$. Witness: E5, 49 exact pairs. □

Assumption F (Causal mediation)**.** The admitted propagation model, or experimental evidence, states that the record laws coincide before a positive time τ.

Without Assumption F nothing above gives a positive delay, and in no case a value in seconds.

9 The sign

Proposition 9.1 (No negative direction from the forced geometry)**.** *Pullbacks, positive sums and restrictions of positive forms are positive:* $(Jv)^\top F(Jv) \geq 0$. *By Sylvester's law of inertia, invertible changes of coordinates preserve signature. The forced geometry cannot produce a negative direction.*

Assumption G (Radial declaration)**.** On an open cone of parameters there is a potential Ψ with positive definite Hessian F and log-homogeneity $\Psi(\lambda z) = \Psi(z) - 2\alpha \log \lambda$, i.e. a declared scale action with a distinguished radial direction.

Theorem 9.2 (Radial reflection)**.** *Under Assumption G, the form* $g = F - \alpha^{-1} d\Psi \otimes d\Psi$ *has inertia* $(1, d-1)$. *The radial direction is negative, and its* F*-orthogonal complement is positive.*

Proof. Euler's identity gives $d\Psi(z)[z] = -2\alpha$. Differentiating gives $F(z, \cdot) = -d\Psi$, hence $F(z,z) = 2\alpha$. Put $u = z/\sqrt{2\alpha}$. Then $g = F - 2F(u,\cdot)^{\otimes 2}$. For $v = au + w$ with $F(u,w) = 0$, we get $g(v,v) = -a^2 + F(w,w)$. Witness: E10, where for $q = t^2 - |x|^2$ and $\Psi = -\frac{3}{2}\log q$ the identity $g = -(\alpha/q)\operatorname{Hess} q$ holds symbolically, $F > 0$, and g has inertia $(1,2)$. □

The positive directions are supplied by the forced geometry. The single negative one is supplied by the declaration. The wall-tap's unit-dilation symmetry is a candidate scale action, but turning it into a log-homogeneous potential is an additional declaration.

10 Obstructions: what tapping does not force

Obstruction 10.1 (Unperformed protocol). Two models can agree on every performed protocol and differ on an unperformed one (E11). No analysis of the performed records selects the response to the unperformed protocol.

Obstruction 10.2 (Finite records). Under i.i.d. Bernoulli runs with $p = \frac{1}{3}$ or $p = \frac{1}{2}$, every finite string has positive probability under both. A finite record supports statistical confirmation, never deductive identification.

Obstruction 10.3 (Mediator). A black box reproducing the same controlled response laws is observationally equivalent under that protocol family. The internal mediator is not identified.

Obstruction 10.4 (Geometry). A finite response table need not admit a smooth parameterization. A supplied one may have a singular or lower-dimensional image.

Obstruction 10.5 (Signature and delay). Positive forms never pull back to Lorentzian ones. A positive arrival time requires Assumption F, and a Lorentz sign requires Assumption G.

11 Back-identification: the established names, after the fact

Derived here	Recognized as
Lemma 2.2: processing and composition	stochastic kernels, Markov categories
Theorem 3.2: descent	identifiability; quotient factorization
Theorem 3.5: predictive update	predictive-state representations; behavioural equivalence
Theorem 4.1: monotone risk	forward Blackwell comparison; data processing
Theorem 4.3: reversible refinements	Čencov's congruent Markov morphisms
Theorem 4.4: converse comparison	Blackwell–Sherman–Stein
Theorem 5.4: topology of confirmation	initial (weak) topology; statistical verifiability (Genin–Kelly)
Theorem 6.3: gluing	Čech H^1 of a cover; graph cohomology; holonomy
Theorem 7.1: forced geometry	Čencov's characterization (Campbell's extension)
Theorem 7.2: non-ideal geometry	stability of the characterization
Theorem 7.3: exact deficit	conditional score variance; Fisher data processing
Hessian selection	Hessian characterization of divergences (Eguchi)
Regular error bound	Cramér–Rao
Confirmation identity	Neyman–Pearson / Le Cam testing; total variation
Theorem 9.2: radial reflection	Koszul–Vinberg cone geometry; hyperbolic polynomials

12 Assumption ledger: ideal and non-ideal

Result	Uses	Ideal → non-ideal
Records, processing	A, B	limits assumed; finite samples via Section 5
Descent, refinement	exact laws	approximate laws: tolerance neighbourhoods (Section 5)
Predictive quotient	A, C	no finite-dimensionality claimed
Monotone risk; reversible refinements	B	exact; approximate simulation measured by deficiency
Converse comparison	admitted separation lemma	—
Topology	A, D, countably many protocols	non-i.i.d. records need their own concentration
Gluing	group-valued conversions, connected graph	noisy conversions: holonomy with uncertainty, so accept / reject / abstain
Forced geometry	E	approximate invariance: $e^{\pm 2\varepsilon}$ (Theorem 7.2)
Deficit, development, error bound	smooth model; unbiasedness for the bound	exact identities
Confirmation delay	F for $\tau > 0$	threshold times stable only with strict crossing
Lorentz sign	G	quadratic only for rank-two cones (companion paper)

What is locked. A declared stable classical intervention–record system determines the following:

- its observational classes and predictive quotient;
- its processing order, and its reversible refinements;
- its topology of confirmation;
- its gluing obstructions.

If a universal continuous positive quadratic geometry must ignore reversible refinements, that geometry is c·Fisher. Its contraction, deficit, Hessian form, error bound, development and confirmation identity follow by direct calculation. A Lorentz sign needs one declared radial structure.

What is not locked. That all physical mathematics follows from one tap. The obstructions of Section 10 show exactly where further structure enters.

13 Witnesses

`rg_walltap_derivation.py` runs E1–E11 in exact rational arithmetic (symbolic for E9–E10; LP feasibility for the left-inverse side of E2). Reruns are byte-identical.

- E1: 81 deficit identities.
- E2: 225 maps, 24 of them recoverable.
- E3–E4: the splitting identities.
- E5: 49 testing identities.
- E6: the hybrid bound on all events.
- E7: the elementary verifier.
- E8: graph gluing.
- E9: the error bound attained.
- E10: the radial inertia.
- E11: the unperformed protocol.

These are finite algebraic witnesses, not substitutes for the proofs or a proof-assistant certificate.